

RTDS Top 3 Query

--Query 1:- Direct connection between two stations

-- E.g, Ahmedabad, figure out nearby stations in same division/zone that still have no direct trains.

```
WITH ahd AS (
  SELECT
    st.station_id    AS ahd_id,
    d.zone_id        AS ahd_zone_id
  FROM station st
  JOIN division d
    ON d.division_id = st.division_id
  WHERE st.station_name = 'Ahmedabad Junction'
  LIMIT 1
),
same_area_stations AS (
  SELECT
    st.station_id,
    st.station_name,
    st.city,
    st.division_id,
    d.zone_id
  FROM station st
  JOIN division d
    ON d.division_id = st.division_id
  CROSS JOIN ahd
  WHERE st.station_id <> ahd.ahd_id
  AND (
    d.zone_id    = ahd.ahd_zone_id
  )
),
direct_pairs AS (
  SELECT DISTINCT
    sch.destination_station_id AS dest_id
  FROM schedule sch
  JOIN ahd
    ON sch.source_station_id = ahd.ahd_id
)
SELECT
  s.station_id,
  s.station_name,
  s.city
FROM same_area_stations s
LEFT JOIN direct_pairs dp
```

ON dp.dest_id = s.station_id
WHERE dp.dest_id IS NULL
ORDER BY s.station_name;

	station_id [PK] character varying (20) 	station_name character varying (120) 	city character varying (60) 
1	ST047	Anand Junction	Anand
2	ST005	Bandra Terminus	Mumbai
3	ST048	Bharuch Junction	Bharuch
4	ST045	Bhavnagar Terminus	Bhavnagar
5	ST044	Jamnagar Railway Stati...	Jamnagar
6	ST042	Neemuch Railway Stati...	Neemuch
7	ST043	Rajkot Junction	Rajkot
8	ST041	Ratlam Junction	Ratlam
9	ST029	Surat	Surat
10	ST030	Vadodara Junction	Vadodara
11	ST046	Veraval Junction	Veraval

--Query 2:- Train Searching

--Returns all train from boarding station to destination station.

```

SELECT
    t.train_id,
    t.train_type,
    tc.class_name           AS coach_class,
    sch.starting_ts         AS departure_time,
    sch.ending_ts          AS arrival_time,
    t.base_fare             AS base_fare,
    r.total_distance        AS route_distance_km
FROM train t
JOIN route r              ON r.route_id = t.route_id
JOIN train_class tc        ON tc.class_id = t.class_id
JOIN schedule sch          ON sch.train_id = t.train_id
                        AND sch.starting_ts > CURRENT_TIMESTAMP
JOIN route_station brd_rs  ON brd_rs.route_id = r.route_id
                        AND brd_rs.station_id = 'ST001'
JOIN route_station dst_rs  ON dst_rs.route_id = r.route_id
                        AND dst_rs.station_id = 'ST004'
                        AND dst_rs.visiting_order > brd_rs.visiting_order

```

WHERE t.train_status = 'Active'
ORDER BY sch.starting_ts;

	train_id character varying (20)	train_type character varying (60)	coach_class character varying (60)	departure_time timestamp without time zone	arrival_time timestamp without time zone	base_fare numeric (10,2)	route_distance_km numeric (10,2)
1	TR001	Rajdhani	AC Second Tier (2A)	2026-04-21 16:30:00	2026-04-22 08:15:00	1550.00	1384.48
2	TR001	Rajdhani	AC Second Tier (2A)	2026-04-22 16:30:00	2026-04-23 08:15:00	1550.00	1384.48
3	TR001	Rajdhani	AC Second Tier (2A)	2026-04-23 16:30:00	2026-04-24 08:15:00	1550.00	1384.48
4	TR001	Rajdhani	AC Second Tier (2A)	2026-04-24 16:30:00	2026-04-25 08:15:00	1550.00	1384.48
5	TR001	Rajdhani	AC Second Tier (2A)	2026-04-25 16:30:00	2026-04-26 08:15:00	1550.00	1384.48
6	TR001	Rajdhani	AC Second Tier (2A)	2026-04-26 16:30:00	2026-04-27 08:15:00	1550.00	1384.48
7	TR001	Rajdhani	AC Second Tier (2A)	2026-04-27 16:30:00	2026-04-28 08:15:00	1550.00	1384.48
8	TR001	Rajdhani	AC Second Tier (2A)	2026-04-28 16:30:00	2026-04-29 08:15:00	1550.00	1384.48

--Query 3:- Ranks all the users

--Groups users by their booking history, counts tickets purchased, sums payment amounts, and orders results descending

```
SELECT
    ru.user_id,
    ru.name,
    ru.email,
    COUNT(tk.pnr) AS total_tickets_booked,
    COALESCE(SUM(p.amount), 0) AS total_revenue
FROM registered_user ru
LEFT JOIN tickets tk
    ON tk.user_id = ru.user_id
LEFT JOIN payment p
    ON p.transaction_id = tk.transaction_id
GROUP BY ru.user_id, ru.name, ru.email
ORDER BY total_revenue DESC, total_tickets_booked DESC;
```

	user_id [PK] character varying (20) 	name character varying (100) 	email character varying (120) 	total_tickets_booked bigint 	total_revenue numeric 
1	U016	Sunil Joshi	sunil.joshi@gmail.com	3	12800.00
2	U040	Neeraj Saxena	neeraj.saxena@gmail.com	3	11110.00
3	U011	Tirth Solanki	tirth.solanki@gmail.com	3	10430.00
4	U004	Rajesh Singh	rajesh.singh@gmail.com	4	10198.00
5	U032	Ajay Bhatia	ajay.bhatia@gmail.com	3	9985.00
6	U028	Karan Ahuja	karan.ahuja@gmail.com	3	9780.00
7	U008	Taksh Chauhan	taksh.chauhan@gmail.com	3	9300.00
8	U046	Hemant Gulati	hemant.gulati@gmail.com	3	8925.00
9	U012	Preet Hirapara	preet.hirapara@gmail.com	3	8725.00
10	U002	Sanjay Gupta	sanjay.gupta@gmail.com	4	8720.00
11	U044	Kavita Kaur	kavita.kaur@gmail.com	3	8710.00
12	U020	Gopal Kumar	gopal.kumar@gmail.com	3	8690.00
13	U047	Zafar Khan	zafar.khan@gmail.com	3	8490.00
14	U007	Dhaval Tolani	dhaval.tolani@gmail.com	3	8430.00
15	U036	Raju Rastogi	raju.rastogi@gmail.com	3	8235.00
16	U001	Ramesh Kumar	ramesh.kumar@gmail.com	4	8205.00
17	U003	Amit Sharma	amit.sharma@gmail.com	4	8020.00
18	U027	Tushar Kapoor	tushar.kapoor@gmail.com	3	7905.00
19	U022	Deepak Reddy	deepak.reddy@gmail.com	3	7820.00
20	U048	Lalit Jain	lalit.jain@gmail.com	3	7640.00
21	U035	Farhan Qureshi	farhan.qureshi@gmail.com	3	7545.00
22	U024	Prakash Nair	prakash.nair@gmail.com	3	7481.00
23	U006	Ronak Vataliya	ronak.vataliya@gmail.com	3	7105.00
24	U023	Uday Shetty	uday.shetty@gmail.com	3	6950.00
25	U031	Shyam Kumar	shyam.kumar@gmail.com	3	6925.00
26	U015	Arvind Desai	arvind.desai@gmail.com	3	6750.00
27	U068	Arjun Bhatia	arjun.bhatia@gmail.com	1	6400.00
28	U064	Vivek Rathi	vivek.rathi@gmail.com	1	6400.00
29	U092	Mohit Chauhan	mohit.chauhan@gmail.com	1	6400.00
30	U072	Kabir Rastogi	kabir.rastogi@gmail.com	1	6400.00
31	U076	Dhruv Saxena	dhruv.saxena@gmail.com	1	6400.00
32	U088	Harsha Vardhan	harsha.vardhan@gmail.com	1	6400.00
33	U060	Rohan Das	rohan.das@gmail.com	1	6400.00
34	U056	Aman Desai	aman.desai@gmail.com	1	6400.00
35	U084	Siddhant Saraf	siddhant.saraf@gmail.com	1	6400.00
36	U100	Rajeev Menon	rajeev.menon@gmail.com	1	6400.00
37	U096	Tarun Varma	tarun.varma@gmail.com	1	6400.00
38	U052	Rahul Verma	rahul.verma@gmail.com	1	6400.00
39	U080	Dev Kaur	dev.kaur@gmail.com	1	6400.00
40	U010	Darpan Sherahiya	darpan.sherahiya@gmail.com	3	6010.00